Skyscraper Stack — Chat Export

This PDF contains the available conversation context for the Skyscraper Stack Pygame project, including the implementation notes for Tasks 1–4.

Task 1 — Fix block placement

The original overlap condition was inverted. Horizontal overlap is calculated as right - left. A positive value means the blocks overlap; zero means they only touch; a negative value means they are separated. The successful condition was changed to **is_successful_drop = overlap > 0**.

Task 2 — Perfect placements

Perfect placement is detected when **abs(act.x - top_block.x) <= 3**. The block snaps into alignment and keeps its width. A golden PERFECT popup lasts 30 frames. Perfect streaks are tracked; after 3 consecutive perfect placements the block expands by 10 pixels total, 5 pixels on each side. Normal placements reset the streak. The implementation gives normal placements +1 score and perfect placements +3.

Task 3 — Falling debris

A Debris class was added. Normal partial-overlap placements create debris from the left and/or right portions hanging outside the tower. Debris has velocity, gravity, rotation, and is removed after falling below the screen. Multiple debris pieces are supported. Perfect placements create no debris.

Task 4 — Atmospheric sky progression

The background changes according to tower height / number of successfully placed blocks, not time. The stages progress from twilight blue to dusk purple to night/starfield to very dark stratosphere/black. Colors transition gradually through linear interpolation. The physical tower height uses **len(self.stack) - 1** because perfect-placement bonus points make score different from actual tower height. Stars are generated once during initialization instead of being randomly generated every frame. The sky is rendered before all game objects, and the existing camera scrolling remains unchanged.

Atmospheric configuration

Sky colors used: twilight (45, 75, 120), dusk (75, 45, 100), night (18, 24, 55), and very dark (5, 7, 18). Each atmospheric transition spans 15 placed blocks. Stars begin fading in after 18 blocks and reach full opacity at 30 blocks.

Current implementation scope

All Tasks 1–4 are localized to **game/game_engine.py**. Task 4 does not modify the Task 1 overlap logic, Task 2 perfect-placement logic, or Task 3 debris behavior.